



Hot Water Treatment and Postharvest Loss in Carabao Mango

Treatment × season effects on physiological and pathological loss

Synthetic data generated by an LLM for training purposes. Not for real-world decisions.

Season

All

Hot Water Treated

All

Year

All

Province

All

Type of Loss

Pathological Loss

Total Batches

48

Starting Weight

15,279 kg

Pathological Loss

977 kg

Physiological Loss

1,005 kg

Est. Gross Savings

₱17,079

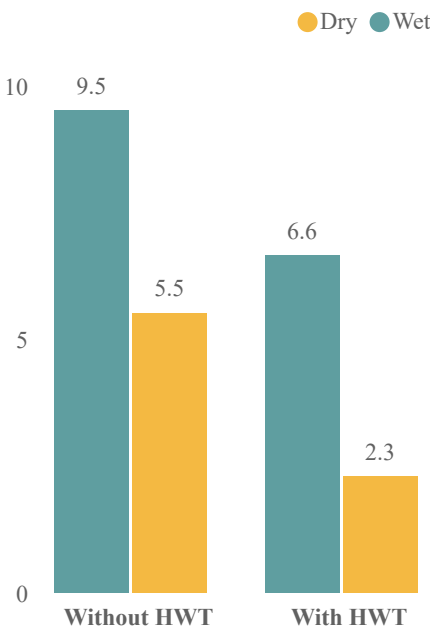
HWT Adoption Rate

33%

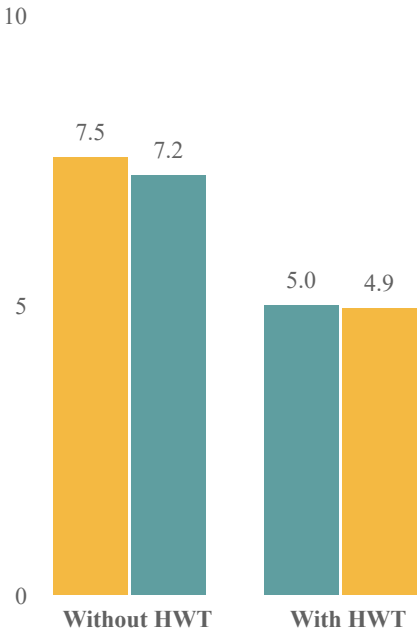
Money KPI Status

✓ Peso figures valid for this view.

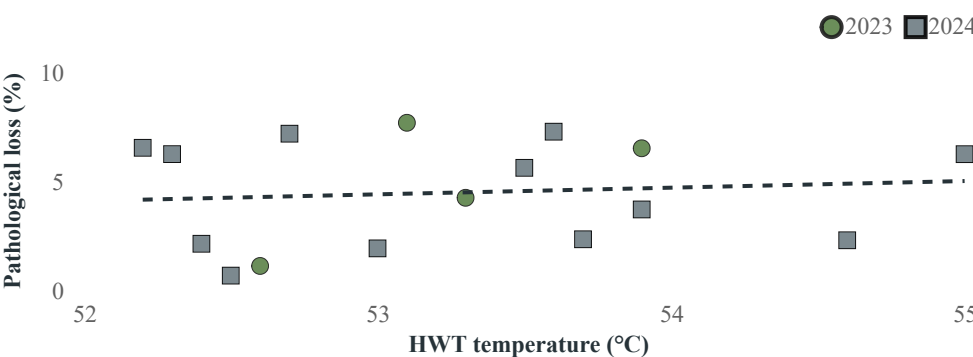
Pathological Loss (% per batch)



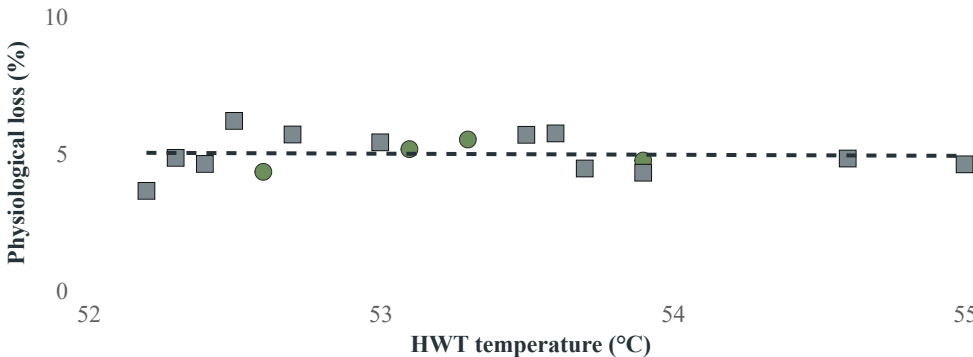
Physiological Loss (% per batch)



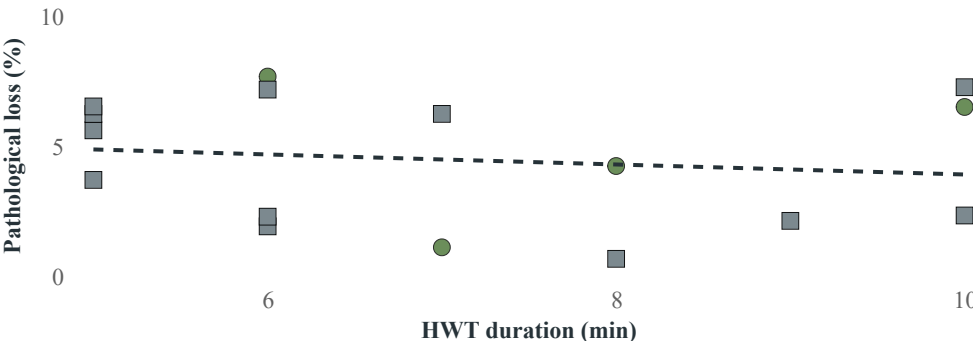
Pathological Loss vs. HWT Temperature



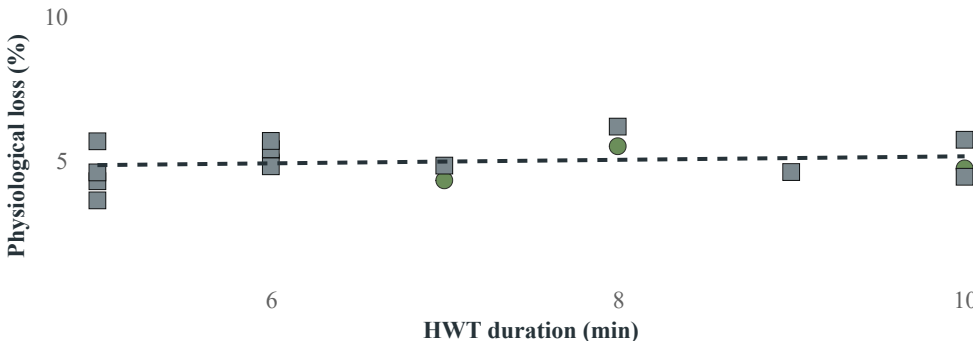
Physiological Loss vs. HWT Temperature



Pathological Loss vs. HWT Duration



Physiological Loss vs. HWT Duration



Key Insights

- HWT results in the reduction of pathological and physiological loss by ~3 and ~2.4 percentage points (pp) per batch, respectively (≈40% and ≈33% relative reduction).
- Season has a significant effect on pathological loss, raising it by ~4 pp, but barely changes physiological loss (< 0.3 pp).
 - Wet season is consistent with increased risk of anthracnose and stem-end rot infection and proliferation under humid conditions.
- The increase in adoption rate (17% → 50%) reduces the loss in general, but the per-batch treatment effect does not change between years

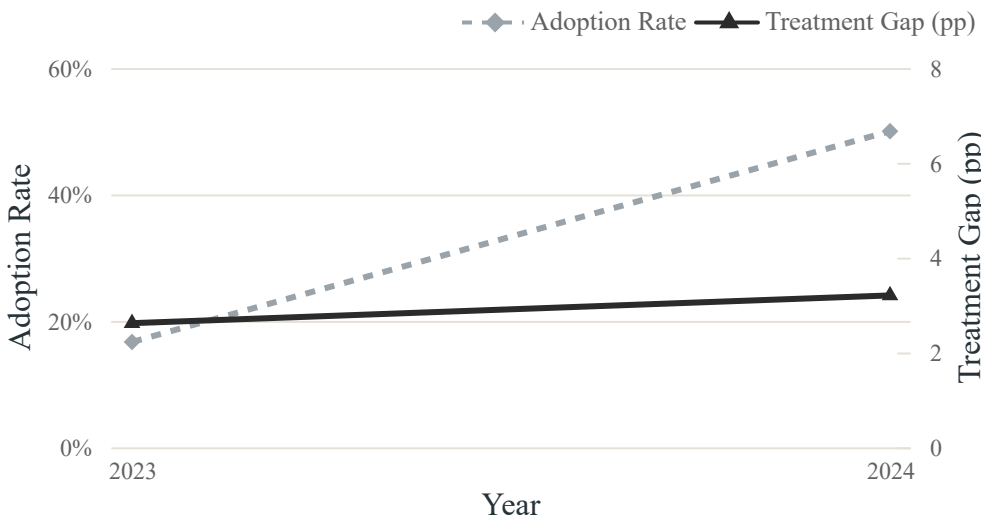
Recommendations

- Promote the use of HWT to further reduce loss, especially in wet season (Jun–Nov).
- Keep treatment within the validated 52–55 °C / 5–10 min window; equipment should hold this range reliably rather than exceed it.
- Measure HWT operating cost per batch to compute net benefit.

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More Batches Treated, Same Effect per Batch

Adoption: 17% → 50% · Treatment gap: ~3 pp in both years



Price and Decay Move Together Only Because of Season

The overall trend comes from wet-season fruit being both pricier and more decay-prone

